1. Scan your local Wi-Fi network with Nmap

Step 1 — Check your IP address

In Kali, run:

- ip addr

```
┌──(root㉿kali)-[~]
└─# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5a:1f:35 brd ff:ff:ff:ff:ff:ff
    inet 192.168.182.138/24 brd 192.168.182.255 scope global dynamic noprefixroute eth0
       valid_lft 1517sec preferred_lft 1517sec
    inet6 fe80::20c:29ff:fe5a:1f35/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Step 2 — Discover connected devices

Run:

- sudo nmap -sn 192.168.1.0/24

```
┌──(root㉿kali)-[~]
└─# nmap -sn 192.168.182.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 10:00 IST
Nmap scan report for 192.168.182.1
Host is up (0.00076s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.182.2
Host is up (0.00046s latency).
MAC Address: 00:50:56:EB:B1:8E (VMware)
Nmap scan report for 192.168.182.254
Host is up (0.00035s latency).
MAC Address: 00:50:56:E0:16:10 (VMware)
Nmap scan report for 192.168.182.138
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 2.16 seconds
```

Step 3 — Record your results

Ip address	Device/Hostname	Status
192.168.182.1	Router	up

192.168.182.138	Kali-linux	up
192.168.182.2	Unknown	up

Task 2: Display the ARP table

Run:

- arp -a

```
┌──(root㉿kali)-[~]
└─# arp -a
? (192.168.182.2) at 00:50:56:eb:b1:8e [ether] on eth0
? (192.168.182.254) at 00:50:56:e0:16:10 [ether] on eth0
```

Task 3: Scan your own computer for common ports

First find your Kali IP:

- ip addr

Suppose your IP is:

- 192.168.182.138

Then run:

- sudo nmap -p 22,80,443 192.168.182.138

```
┌──(root㉿kali)-[~]
└─# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5a:1f:35 brd ff:ff:ff:ff:ff:ff
    inet 192.168.182.138/24 brd 192.168.182.255 scope global dynamic noprefixroute eth0
       valid_lft 983sec preferred_lft 983sec
    inet6 fe80::20c:29ff:fe5a:1f35/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(root㉿kali)-[~]
└─# nmap -p 22,80,443 192.168.182.138
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-22 10:21 IST
Nmap scan report for 192.168.182.138
Host is up (0.000049s latency).

PORT    STATE  SERVICE
22/tcp  closed ssh
80/tcp  closed http
443/tcp closed https

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
```

Task 4: Scan scanme.nmap.org

Run:

- nmap scanme.nmap.org

Port	Typical service	Purpose
22	SSH	Secure remote administration
80	HTTP	Web traffic
443	HTTPS	Encrypted web traffic
9929	nping-echo	Nmap's Nping echo service
31337	Verious/unknown	Historically associated with Back Orifice; don't assume the service without identification

5. Show only open ports and services

Command is:

- nmap -sV --open 192.168.182.138